

I used Bidirectional LSTM

Using the jigsaw dataset not original data set here are my results

F1- score of 0.84

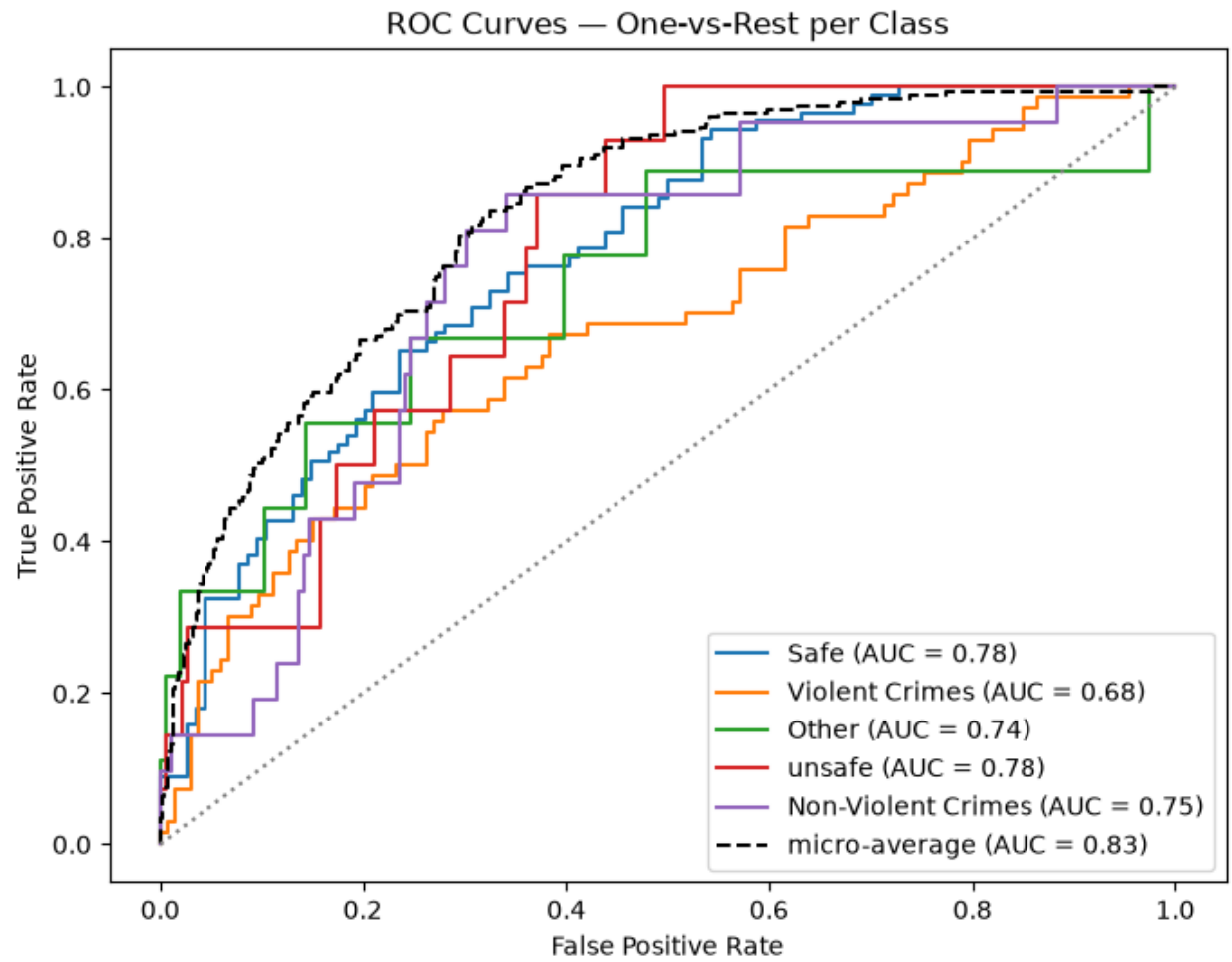
	precision	recall	f1-score	support
Not Toxic	0.97	0.98	0.97	903
Toxic	0.75	0.68	0.71	97
accuracy			0.95	1000
macro avg	0.86	0.83	0.84	1000
weighted avg	0.95	0.95	0.95	1000

	Not Toxic	Toxic
Not Toxic	881	22
Toxic	31	66

Confusion matrix

Using original cellula data here are the results

	precision	recall	f1-score	support
Safe	0.66	0.66	0.66	89
Violent Crimes	0.51	0.53	0.52	70
Other	0.27	0.33	0.30	9
unsafe	0.33	0.29	0.31	14
Non-Violent Crimes	0.18	0.14	0.16	21
accuracy			0.52	203
macro avg	0.39	0.39	0.39	203
weighted avg	0.52	0.52	0.52	203



	Safe	Violent Crimes	Other	unsafe	Non-Violent Crimes
Safe	59	16	6	1	7
Violent Crimes	22	37	1	4	6
Other	1	3	3	1	1
unsafe	0	10	0	4	0
Non-Violent Crimes	8	7	1	2	3

Now for the RRN

Using BI- directional RNN here are the results

The F1 score is 0.8

	precision	recall	f1-score	support
Not Toxic	0.97	0.95	0.96	903
Toxic	0.60	0.69	0.64	97
accuracy			0.93	1000
macro avg	0.78	0.82	0.80	1000
weighted avg	0.93	0.93	0.93	1000

Confusion matrix

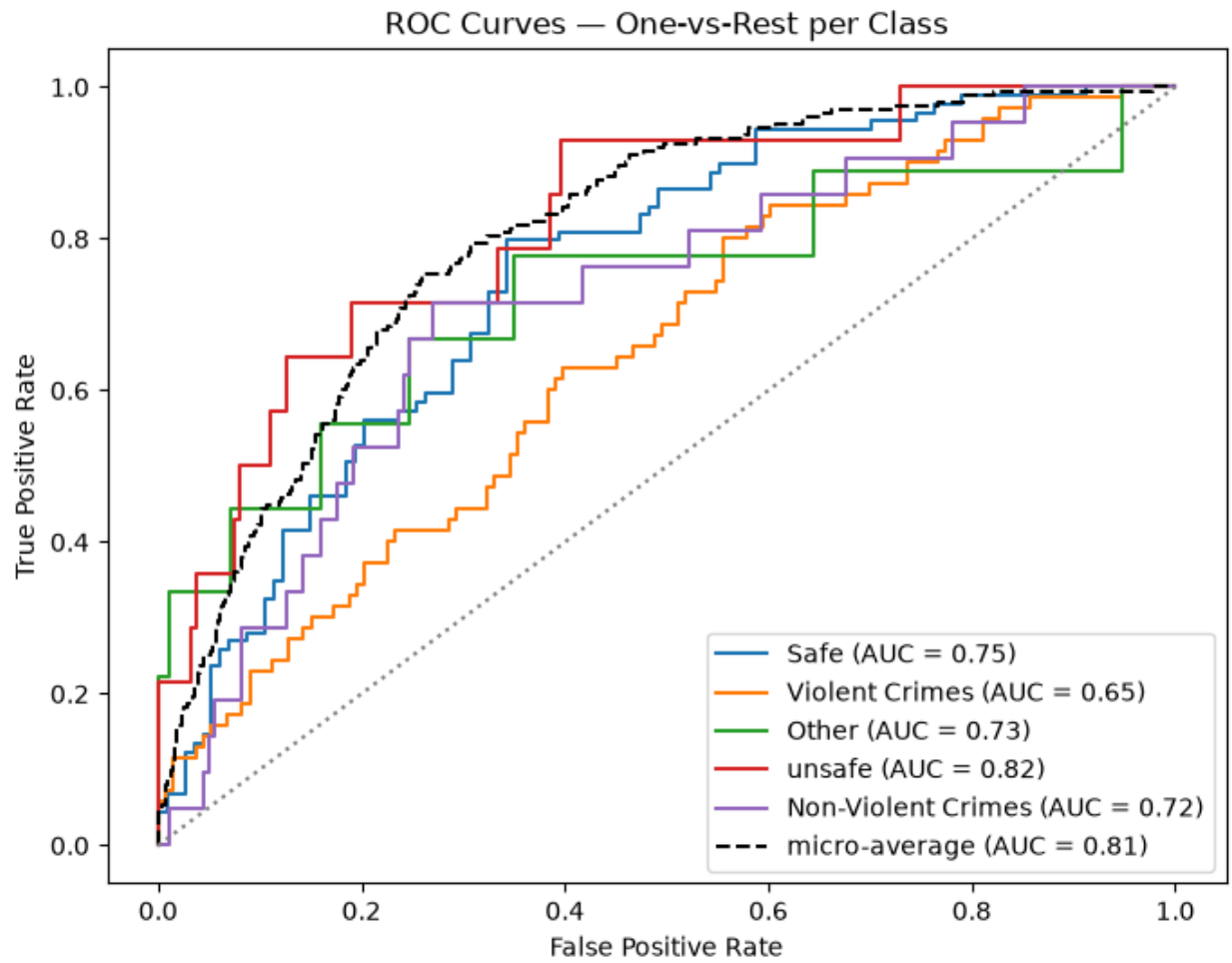
	Not Toxic	Toxic
Not Toxic	859	44
Toxic	30	67

Using original cellula data here are the results

	precision	recall	f1-score	support
Safe	0.67	0.62	0.64	89
Violent Crimes	0.45	0.47	0.46	70
Other	0.30	0.33	0.32	9
unsafe	0.28	0.36	0.31	14
Non-Violent Crimes	0.30	0.29	0.29	21
accuracy			0.50	203
macro avg	0.40	0.41	0.41	203
weighted avg	0.51	0.50	0.51	203

Macro F1 score of 0.41

ROC curve



Confusion matrix

	Safe	Violent Crimes	Other	unsafe	Non-Violent Crimes
Safe	55	21	5	5	3
Violent Crimes	19	33	2	6	10
Other	1	2	3	2	1
unsafe	0	9	0	5	0
Non-Violent Crimes	7	8	0	0	6